

# yolo\_ros

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Provides a ROS package based on PyTorch-YOLO.

## Running Environment:

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- Ubuntu 20.04
- ROS Noetic
- Python >= 3.7.0, PyTorch >= 1.7

## Environment Configuration:

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1. First, install the environment with the corresponding Python and PyTorch versions.
2. Then install the following dependencies:

```
pip install ultralytics  
pip install rospkg
```

3. Install Yolo\_ROS:

```
cd catkin_ws/src  
git clone https://github.com/HT-hlf/yolo_ros.git  
cd ..  
catkin_make
```

## Usage

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### Use YOLO to Detect Objects in Images

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#### Use in Simulation Environment

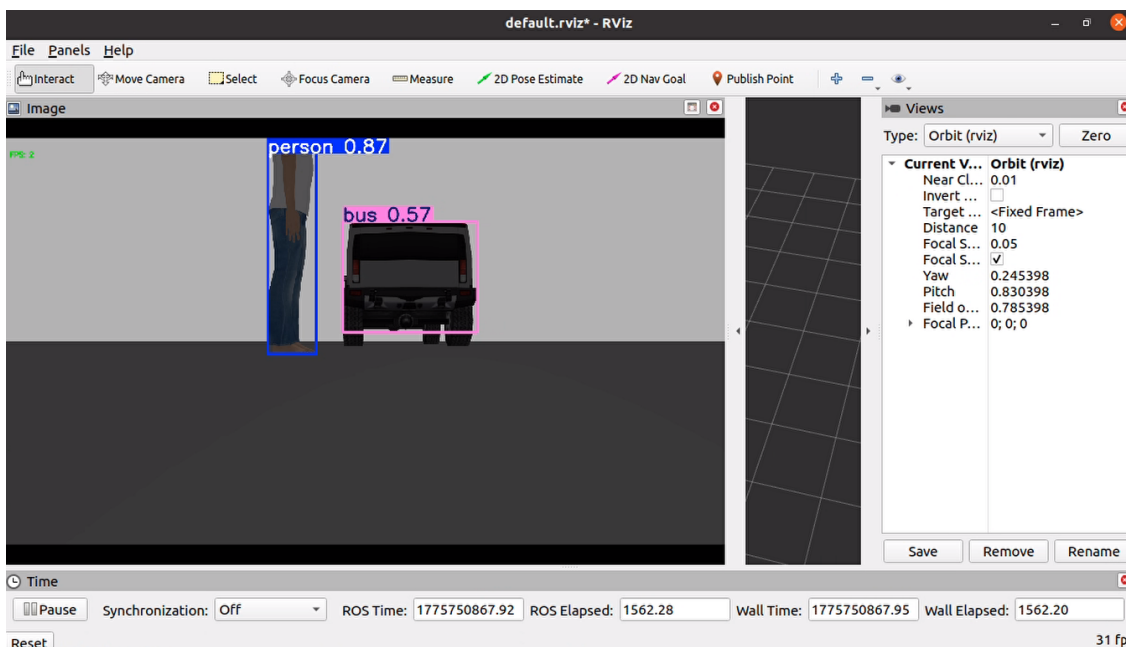
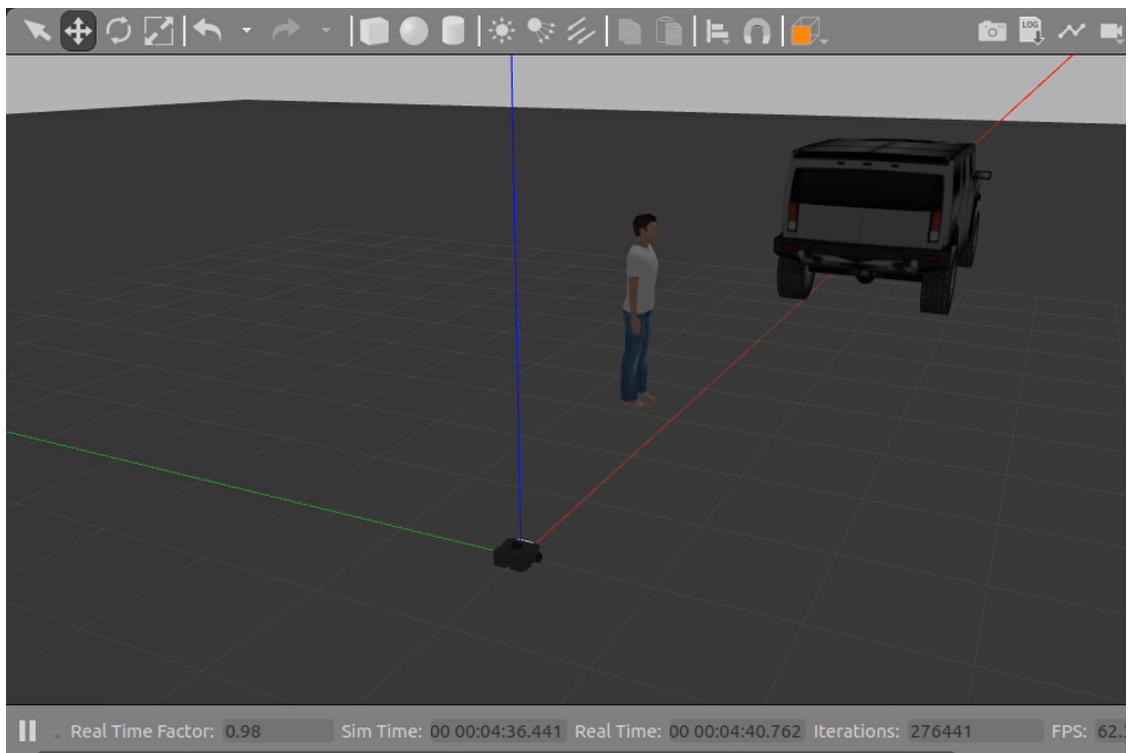
- Start the simulation environment and robot:

```
source devel/setup.bash  
roslaunch turtlebot3_gazebo turtlebot3_empty_world.launch
```

- Start the ros\_yolo node:

```
source devel/setup.bash  
roslaunch yolo_ros yolo.launch
```

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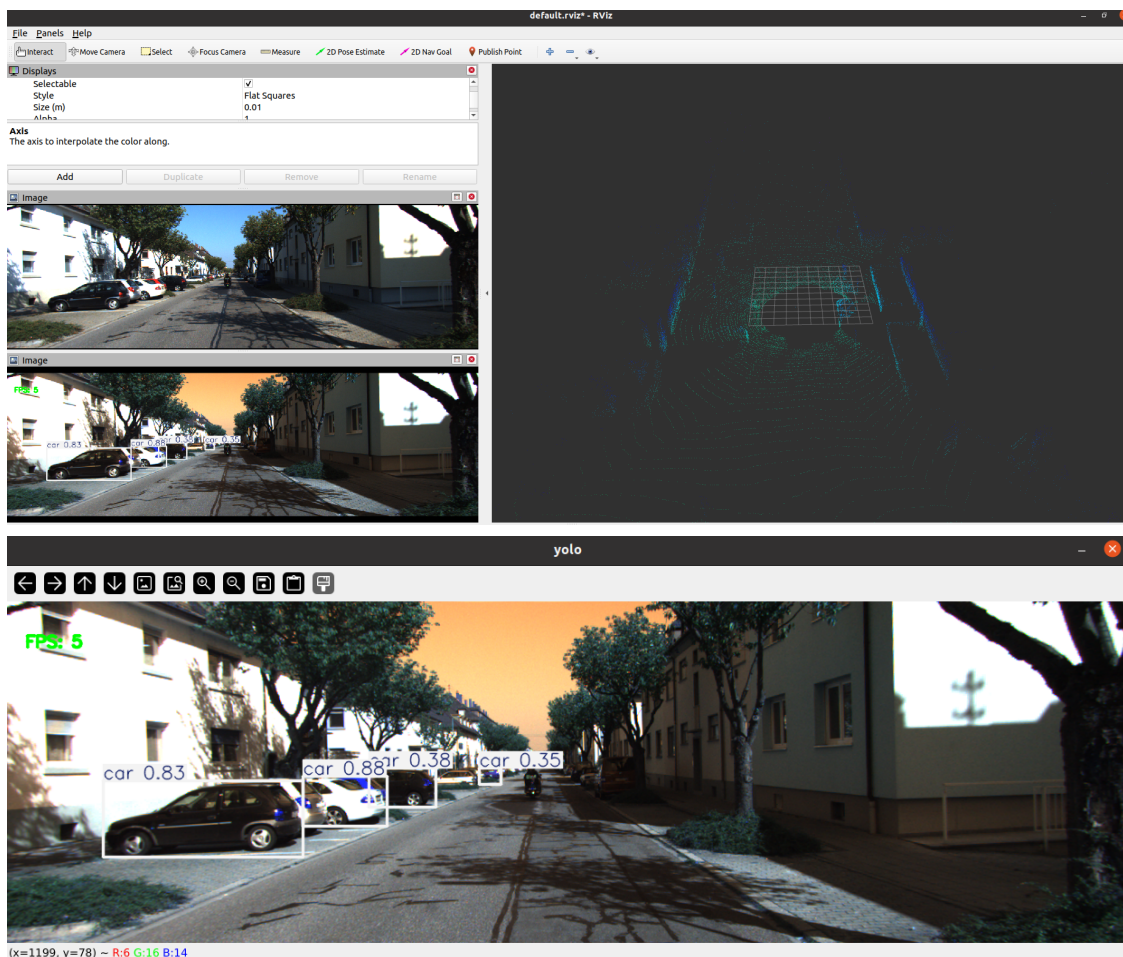
## Use with Recorded Dataset

- Play bag data:

```
roslaunch yolo_ros yolo_kitti.launch
```

- Start the ros\_yolo node:

```
source devel/setup.bash
roslaunch yolo_ros yolo_kitti.launch
```



## Use YOLO to Detect Objects in Images and Calculate Their 3D Coordinates

- Start the simulation environment and robot:

```
source devel/setup.bash
roslaunch turtlebot3_gazebo turtlebot3_empty_world.launch
```

- Start the ros\_yolo3D node:

```
source devel/setup.bash
roslaunch yolo_ros yolo_3D.launch
```

